

ORF307: Optimization

Precept: #6

Agenda

- LP reformulations
- Portfolio least squares

Reminders and Announcements

- Always check Python notebooks in Companion (in directories `lectures/` and `precepts/`)
- Write down the deadline of the current HW:
- Other announcements on blackboard.

Formulate the following problem as an LP

$$\min_x \sum_{i=1}^m \max\{0, a_i^T x + b_i\} \quad (1)$$

Solution:

Portfolio optimization problem

- Current portfolio: $h^{curr} \in \mathbb{R}^n$ is the amount invested in n assets
- Current total value: $\mathbf{1}^T h^{curr}$
- $h - h^{curr}$ is the trade vector
- The n assets are divided into m industry sectors (e.g., financial, pharmaceuticals, semiconductors, etc.)
- $s \in \mathbb{R}^m$ denotes the dollar value sector exposure. We can write

$$s = Sh, \quad \text{where } S \in \mathbb{R}^{m \times n} \quad \text{and} \quad S_{ij} = \begin{cases} 1 & \text{if asst } j \text{ is in sector } i \\ 0 & \text{otherwise} \end{cases}$$

- The new portfolio must have a given (desired) sector exposure described by the vector $s^{des} \in \mathbb{R}^m$
- **Task #1:** minimize the trading cost given by:

$$\sum_{i=1}^n \kappa_i (h_i - h_i^{curr})^2$$

where $\kappa_i \geq 0$, $i = 1, \dots, n$ are some weights related to the trading cost of asset i .

- **Task #2:** Explain how to find h using constrained least squares.

Solution:

Penalize shorting assets

Suppose you want to include a penalty on shorting assets. How might you include this as a linear constraint?

Solution: